Executive Summary

Introduction

In a world characterized by rapid technological progress and a growing number of financial transactions, the task of identifying fraudulent activities within the realm of financial transactions has become increasingly complex and pressing. This project report delves into the ever-evolving intersection of fraud detection and machine learning, addressing the crucial need for accurate and efficient fraud detection methodologies. By focusing on bridging the gap between state-of-the-art data science methodologies and ethical considerations, this research endeavors to offer a comprehensive solution to a problem that profoundly affects businesses, consumers, and the broader economy.

Problem Statement

Financial fraud presents a significant risk to individuals, businesses, and the overall economy. The ever-evolving fraudulent techniques have emphasized the limitations of traditional rule-based methods in detecting fraud. Additionally, the ethical aspect of using automated models, which can inadvertently perpetuate biases, calls for a thoughtful and deliberate approach. This situation poses a crucial challenge: the development of advanced and morally upright fraud detection systems. These systems must not only accurately identify fraudulent activities but also uphold fairness and transparency in their predictions. Further complicating matters is the lack of comprehensive real-world datasets that cover a wide range of scenarios, an essential factor for constructing and training effective detection models.

Research Questions and Significance

The primary focus of this analysis is to thoroughly explore the following primary research inquiries, ultimately shedding light on crucial insights that can shape the advancement of future fraud detection models.

Comparative Performance of Machine Learning Algorithms

The main objective of this study is to examine and compare the performance of four prominent machine learning algorithms: XGBoost, Support Vector Machines (SVM), MLP Classifier, and Multilayer Neural Network with TensorFlow/Keras. The central question in this regard is: among these algorithms, which one demonstrates the most robust and effective performance in identifying instances of fraudulent transactions within the BAF dataset? To address this question, we will utilize evaluation metrics, such as Area Under the Curve (AUC), Precision, and Accuracy. These metrics serve as crucial indicators of the efficacy of each algorithm in distinguishing between legitimate and fraudulent transactions.

Deriving Insights and Future Contributions

Beyond merely comparing algorithmic performance, this analysis aims to uncover noteworthy insights from the comparative analysis of these machine learning techniques in the context of fraud detection using the BAF dataset. These insights are expected to provide valuable guidance for the evolution of future fraud detection models. The second research question revolves around these insights: What significant takeaways can we gain from a comprehensive examination of these machine learning algorithms, and how can these insights influence the trajectory of advancements in fraud detection models?

The implications of this study extend beyond solely assessing algorithms. By addressing these research questions, this analysis seeks to enrich the field of fraud detection with actionable insights that have the potential to shape the development of more accurate, efficient, and ethically responsible fraud detection systems. Through this exploration, a bridge is established between the current assessment and the future direction of research in the realm of financial fraud detection.

Approach and Findings

In the quest for answers, I embarked on a thorough examination of relevant literature, unearthing the immense potential of machine learning and deep learning techniques in the realm of fraud detection. This helped identify pivotal algorithms such as Support Vector Machines (SVM), XGBoost, and neural networks, which possess the ability to unravel intricate patterns and handle vast volumes of data.

The literature review also shed light on the ethical aspects, emphasizing the significance of fairness-aware algorithms to mitigate biases. Concurrently, I delved into the BAF dataset suite

(Jesus et al., 2022) specifically designed for detecting bank account fraud, which has become a highly valuable resource embraced by numerous studies.

This research approach was comprehensive and hands-on. This involved conducting a thorough comparative analysis using various techniques such as undersampling to address biases (Debashree Devi et al., 2020), one-hot encoding for data preprocessing, and separate training and testing sets for accurate assessment.

Next steps included training and testing four prominent algorithms, namely XGBoost, SVM, MLP Classifier, and Multilayer Neural Network with TensorFlow/Keras. Their performance was evaluated using accuracy and AUC metrics, providing insights into correctness and overall effectiveness.

This approach was pragmatic, ensuring that the findings were not merely theoretical but also practically applicable. This in-depth analysis provides actionable insights for future fraud detection efforts, surpassing the realm of data and offering valuable guidance for practical implementation.

An intriguing discovery emerged from this study: the Multilayer Neural Network with TensorFlow/Keras showcased exceptional performance, achieving an accuracy rate of 79.3%. This highlighted the immense potential of deep learning in tackling intricate fraud detection challenges, particularly within the scope of the BAF dataset.

These invaluable insights transcend mere statistics, as they met my objectives of advancing fraud detection methodologies while also ensuring ethical integrity. Through empirical evidence, this analysis serves as a guiding beacon for future models, paving the way for enhanced accuracy and ethical standards.

Implications and Business case

The implications and business case of this research hold significant importance for various stakeholders. For businesses and financial institutions, the implementation of advanced fraud detection systems can effectively mitigate financial losses, safeguard customer data, and uphold trust. It is not only a moral obligation but also a crucial step to protect these organizations from reputational damage and legal consequences.

From a societal standpoint, this research underlines the necessity of ensuring unbiased and fair algorithms. This is vital not only for maintaining social harmony but also for fostering consumer trust in the financial system. As fraud detection systems become increasingly integrated into our daily lives, assuring consumers that their transactions are secure and free from biases becomes of utmost importance.

Conclusion

In conclusion, this project report emphasizes the crucial intersection of fraud detection, machine learning, and ethics. It tackles the challenges of improving the accuracy of fraud detection, promoting fairness, and ensuring transparency. By highlighting the advancements in algorithms and the importance of considering ethical aspects, this research offers guidance for businesses, financial institutions, and policymakers to develop and implement fraud detection systems that are not only effective but also socially responsible. Ultimately, this research serves as a rallying cry, urging stakeholders to embrace ethical data science methodologies for a more secure and just financial landscape.

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Enhancing Bank Account Security: A Comprehensive Study on Machine Learning-Based Fraud Detection

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Chapter 1: Introduction

Financial fraud and security breaches present significant challenges to the integrity of financial systems, undermining customer trust and jeopardizing overall economic stability. The detection and prevention of fraudulent activities necessitate robust and adaptable methodologies, which, in turn, rely on the availability of diverse and relevant data for rigorous testing and validation. Addressing this critical need, the Bank Account Fraud (BAF) Dataset Suite offers synthetic data that closely emulates real-world financial transactions while ensuring the confidentiality of individuals and organizations involved. Serving as a comprehensive collection of synthetic financial transaction data sets, the BAF Dataset Suite aims to foster research and innovation in the realm of fraud detection and financial security.

In this project, my aim is to conduct a comparative analysis of various machine learning algorithms to detect online bank account fraud using the BAF dataset suite. The BAF dataset suite, which was introduced at NeurIPS 2022, tackles the unique challenges of detecting bank account fraud. This comprehensive suite consists of realistic tabular data that accurately represents various transactional scenarios encountered in real-world banking operations. By providing standardized data, the BAF dataset suite enables an accurate evaluation of the performance of machine learning algorithms in fraud detection.

The algorithms selected for analysis include Support Vector Machine (SVM), XGBoost, Multi-Layer Perceptron (MLP) classifier, and Multilayer Neural Network TensorFlow/Keras. By evaluating the performance of these algorithms on the BAF dataset suite, I can gain valuable insights into their strengths and weaknesses in identifying fraudulent transactions.

To present my findings and analysis, I have chosen to undertake a dissertation-style project. This format enables me to deeply explore the subject matter, conduct meticulous research, and deliver a well-organized and comprehensive examination of the performance of the selected machine learning algorithms. This project aims to make significant contributions to the advancement of fraud detection techniques, offering invaluable insights for financial institutions and researchers striving to effectively combat bank account fraud.

1.1. Context & Motivation

In the current digital landscape, the prevalence of online operations, encompassing payments, transactions, and account openings, has fundamentally transformed convenience and accessibility (Hosein, 2009). However, this paradigm shift has led to a simultaneous surge in fraudulent activities, posing substantial financial risks. The significance and relevance of the research topic cannot be understated in this context. Nowadays, digital interactions have become

the norm, rendering fraud attempts increasingly sophisticated and widespread. Considering this, the research focuses on the crucial juncture of account opening, recognizing it as the primary line of defense against potentially fraudulent activities. Left unchecked, these accounts can become breeding grounds for substantial financial losses, making early detection a paramount necessity.

The contemporary digital landscape presents both opportunities and challenges, acting as a sword that cuts both ways. As payment methods and channels rapidly evolve, criminals are equally equipped with innovative fraud techniques (Alkhalil et al., 2021). To tackle this ever-evolving threat, this research delves into advanced fraud detection solutions. Importantly, the trust of customers is at stake. Instances of fraud erode consumer confidence in online transactions, hindering the sustained growth of digital activities. Therefore, this research emerges as a shield, ensuring the establishment of robust fraud detection mechanisms to safeguard customer trust.

The relevance and significance of the research lie in its potential to prevent fraudulent activities during the account opening process. This not only avoids negative financial consequences, but also enhances customer confidence. Multiple stakeholders, including financial institutions, e-commerce businesses, consumers, and regulatory bodies, stand to benefit from this study. Ultimately, this research sheds light on the path towards a more secure digital landscape, fostering the integrity of transactions and instilling trust in digital interactions.

1.2. Research Questions & Objectives

In the rapidly evolving landscape of digital financial transactions, the persistent challenge of detecting and preventing fraudulent activities has become increasingly complex and critical. While various machine learning algorithms have been applied to fraud detection, there remains a gap in comprehensive evaluations of these algorithms specifically tailored for bank account fraud detection. Furthermore, there is a notable absence of comparative studies that assess the performance of these algorithms in the context of bank account fraud detection.

To address these critical research gaps, this study aims to achieve several interconnected questions:

1. Comprehensive Algorithm Evaluation: The primary objective of this research is to conduct a thorough and systematic evaluation of the performance of four distinct machine learning algorithms: Support Vector Machines (SVM), XGBoost, Multi-Layer Perceptron (MLP) classifier, and Multilayer Neural Network with TensorFlow/Keras. The main intention is to assess these algorithms' ability to accurately identify instances of fraudulent bank account

openings using the BAF dataset, a comprehensive and realistic tabular dataset expressly designed for bank account fraud detection research. Through this evaluation, we aim to provide quantitative insights into the strengths and weaknesses of each algorithm.

2. Addressing the Research Gap: This study endeavors to fill a significant research gap by delivering a comparative analysis of these machine learning algorithms within the specific context of bank account fraud detection. Systematically benchmarking these algorithms against one another seeks to contribute essential knowledge regarding which algorithm exhibits the most robust performance. This analysis will be grounded in essential evaluation metrics, including Area Under the Curve (AUC), Precision, and Accuracy.

3. Deriving Valuable Insights: Beyond algorithmic performance, this research seeks to extract valuable insights from the comparative analysis. The aim is to identify trends, patterns, and considerations that can guide the development of future fraud detection models. Answering the question of what can be learned from this comprehensive examination, provide actionable guidance for the evolution of fraud detection techniques and technologies.

4. Contributing to Fraud Detection Research: This study is positioned to make a substantial contribution to the broader field of fraud detection research. By addressing the specific challenges of bank account fraud detection and conducting a comparative analysis of machine learning algorithms, aims to advance the state-of-the-art in fraud detection. This research carries significance not only for financial institutions but also for e-commerce businesses, consumers, and regulatory bodies, all of whom benefit from improved fraud detection mechanisms.

Chapter 2: Literature Review

Fraud detection in the realm of financial transactions presents an ongoing challenge, necessitating the integration of cutting-edge data science methods and ethical considerations. Over the last decade, significant strides have been made by leveraging machine learning algorithms to improve the accuracy and efficiency of fraud detection. This section of the report delves into the existing literature, providing a comprehensive overview of contributions that have greatly advanced fraud detection methodologies. Furthermore, it explores the role of the BAF dataset suite and its valuable contributions to the evolution of fraud detection research.

2.1. Fraud Detection Methodologies

2.1.1. Machine Learning Approaches

Machine learning has emerged as a powerful tool in detecting fraudulent activities due to its ability to process vast amounts of data and identify subtle patterns. This section provides an overview of key machine learning approaches in fraud detection:

A groundbreaking study conducted by Dal Pozzolo et al. (2014) showcased the effectiveness of Support Vector Machines (SVM) in fraud detection. The researchers emphasized the significance of feature engineering and model evaluation techniques.

The XGBoost algorithm, known for its ensemble nature, has garnered considerable attention in handling imbalanced datasets (Chen & Guestrin, 2016). In an extensive study conducted by Lei et al. (2020), it was further highlighted that XGBoost provides outstanding versatility in managing large-scale data. Additionally, the introduction of XGBoost, as outlined in the paper "XGBoost: A Scalable Tree Boosting System," offers scalable solutions for various domains, including fraud detection. Its efficiency and scalability position it as a potent tool for handling large-scale datasets.

In Chris Oakley's enlightening article titled "The Role of Machine Learning in Fraud Detection," the author presents a comprehensive overview of the vital significance that machine learning (ML) models have in the field of fraud detection. ML has brought about a revolution, completely transforming the way businesses, banks, and financial institutions combat fraudulent activities by enabling swift and efficient identification of suspicious transactions on a large scale (Oakley, 2021).

Similarly, the paper "Machine Learning for Fraud Detection: Fundamentals and Benefits" delves deeper into the core principles of applying machine learning to fraud detection. The paper highlights the critical role of data input, as well as the identification and extraction of features, in training machine learning models effectively (Ranjan, 2020). Through the analysis of historical data, these models can distinguish between normal and anomalous behaviors, enabling real-time decision-making regarding the probability of fraudulent activities (Ranjan, 2020).

2.1.2. Neural Networks and Deep Learning Approaches

The emergence of neural networks and the advancements in deep learning have significantly transformed the capabilities of fraud detection. In a study conducted by Bhattacharyya et al. (2019), the effectiveness of Multi-Layer Perceptron (MLP) architectures in capturing intricate relationships within data was showcased. The adaptability of MLPs to complex data structures positions them as an ideal choice for fraud detection. Furthermore, deep learning techniques such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) enable the analysis of sequential data as well as image-based data, as highlighted by Khan et al. (2021).

This groundbreaking technology not only enhances the ability to detect fraudulent activities but also opens new avenues for exploring and understanding data in more comprehensive and detailed ways.

2.1.3. Ethical Considerations & Bias Mitigation

Though there have been advancements in fraud detection techniques, it is essential to acknowledge the ethical concerns and potential biases associated with these models. Recent research by Shetty et al. (2019) has highlighted that biased models can worsen societal inequalities For instance, if a model is trained predominantly on historical data that contains biases, such as racial or gender disparities, it may inadvertently perpetuate these biases by making unfair decisions. This could result in certain groups being disproportionately targeted or disadvantaged in fraud detection processes, exacerbating societal inequalities. This underscores the need to prioritize fairness-aware algorithms and implement bias mitigation strategies to foster the development of responsible fraud detection systems (Dwork et al., 2018).

2.2. Related Works in Fraud Detection

Introducing a specialized bank payments simulator designed for fraud detection research, the paper of (BANKSIM, 2019) offers a realistic and comprehensive collection of tabular data tailored specifically for simulating bank account transactions. By providing standardized and representative data for analysis, this simulator assists researchers in evaluating and comparing the performance of various fraud detection algorithms. An IEEE paper based on BANKSIM highlights the critical role of data input, as well as the identification and extraction of features, in training machine learning models effectively (Ranjan, 2020). Through the analysis of historical data, these models can distinguish between normal and anomalous behaviors, enabling real-time decision-making regarding the probability of fraudulent activities (Ranjan, 2020).

2.3. Research Gap and Objectives

While there exists substantial literature on the application of machine learning and deep learning in fraud detection, a significant research gap exists in terms of a comprehensive comparative analysis of these approaches, particularly in the context of bank account fraud detection. Additionally, the existing research often focuses on specific machine learning algorithms or limited datasets, resulting in a lack of comprehensive comparisons and evaluations across different techniques.

The objective of this research is to address these research gaps by conducting a comparative analysis of four prominent machine learning algorithms, namely XGBoost, Support Vector Machines (SVM), MLP Classifier, and Multilayer Neural Network with TensorFlow/Keras, for the detection of online bank account fraud using the BAF dataset suite. Through this analysis, the

aim is to determine which algorithm demonstrates the most robust performance and to derive valuable insights that can contribute to the advancement of future fraud detection models. Importantly, this comparative analysis will provide a clearer understanding of the strengths and weaknesses of each approach and aid in making informed decisions about the most suitable algorithm for online bank account fraud detection.

Comparing these specific four approaches is important because they represent a diverse set of machine learning techniques commonly used in fraud detection. Each approach has its unique strengths and weaknesses. XGBoost is known for handling imbalanced datasets effectively, SVMs excel in separating complex data, MLP classifiers capture intricate relationships, and custom neural networks offer flexibility. By comparing these, we gain insights into which type of approach is most suitable for the specific challenges of bank account fraud detection, guiding future model development in this critical area.

Chapter 3: Research Methodology

The methodology section of this research project report presents a thorough and systematic approach to compare various machine learning models for fraud detection using the BAF dataset. I have carefully chosen four machine learning models: XGBoost, Support Vector Machines (SVM), MLP Classifier from the Sklearn library, and a custom Multilayer Neural Network implemented using TensorFlow/Keras. Each of these models possesses unique strengths and has been extensively utilized in fraud detection research. This section offers a comprehensive overview of the data preprocessing steps, justification for model selection, implementation details, process of hyperparameter tuning and evaluation metrics for model performance.

3.1. Data Preprocessing

Before implementing machine learning models, I had to preprocess the BAF dataset to ensure its quality and compatibility. First, I imported important Python libraries including pandas, numpy, matplotlib, scikit-learn, xgboost, seaborn, tensorflow, keras, and imbalanced-learn. These libraries played a crucial role in manipulating and visualizing the data, handling imbalanced datasets, and constructing accurate machine learning models. Data preprocessing is a fundamental and essential step in the preparation of a dataset for modeling, as it plays a pivotal role in ensuring the accuracy and reliability of the results. This vital process encompasses various techniques, such as handling missing values, encoding categorical variables, and scaling numerical features.

Next, I loaded the dataset from the file 'Base.csv' into a pandas DataFrame called 'df.' To improve the dataset's quality, I decided to exclude the 'device_fraud_count' column as it didn't provide

any valuable insights since it contained all zeros. Additionally, I carefully examined the dataset and found that there were no missing values, so there was no need for data imputation or removing records. This ensured that the dataset remained intact and complete.

Data Preprocessing

```
print(df['device_fraud_count'].value_counts()) # It's 0 for all rows
df = df.drop(['device_fraud_count'], axis=1, errors='ignore')
print(df)

# to remove null values

df.dropna(inplace=True)
print(df)
```

Figure 1: Code snippet showing data preprocessing

3.2. Methodology for Analysis

Summary statistics were calculated for the dataset's numerical columns using the 'describe()' method, and a bar plot was used to visualize the 'fraud_bool' distribution. Univariate analysis included bar plots for 'income' and payment type frequencies. Bivariate analysis utilized box plots to compare 'income' distributions between fraud and non-fraud transactions. Time-series analysis revealed monthly trends in fraudulent transactions. Data imbalance analysis showed significant class imbalances, leading to the implementation of undersampling techniques.

Furthermore, an interactive Tableau dashboard was created for a comprehensive exploratory data analysis (EDA). It includes visuals on the number of fraud accounts by payment type, class distribution, monthly fraud counts, and customer age frequency. Users can apply filters to explore fraud, non-fraud, or both, enhancing data exploration.

Variables	Description
Income (numeric)	Annual income of the applicant (in decile form). Ranges between [0.1, 0.9]
name_email_similarity (numeric)	Metric of similarity between email and applicant's name. Higher values represent higher similarity. Ranges between [0, 1].
prev_address_months_count (numeric)	Number of months in previous registered address of the applicant, i.e. the applicant's previous residence, if applicable. Ranges between [−1, 380] months (-1 is a missing value).
current_address_months_count (numeric)	Months in currently registered address of the applicant. Ranges between [−1, 429] months (-1 is a missing value).

customer_age (numeric)	Applicant's age in years, rounded to the decade. Ranges between [10, 90] years.
days_since_request (numeric)	Number of days passed since application was done. Ranges between [0, 79] days.
intended_balcon_amount (numeric)	Initial transferred amount for application. Ranges between [−16, 114] (negatives are missing values).
payment_type (categorical)	Credit payment plan type. 5 possible (annonymized) values.
zip_count_4w (numeric)	Number of applications within same zip code in last 4 weeks.Ranges between [1, 6830].
velocity_6h (numeric)	Velocity of total applications made in last 6 hours i.e., average number of applications per hour in the last 6 hours. Ranges between [−175, 16818].
velocity_24h (numeric)	Velocity of total applications made in last 24 hours i.e., average number of applications per hour in the last 24 hours. Ranges between [1297, 9586]
velocity_4w (numeric)	Velocity of total applications made in last 4 weeks, i.e., average number of applications per hour in the last 4 weeks. Ranges between [2825, 7020].
bank_branch_count_8w (numeric)	Number of total applications in the selected bank branch in last 8 weeks. Ranges between [0, 2404]
date_of_birth_distinct_emails_4w (numeric)	Number of emails for applicants with same date of birth in last 4 weeks. Ranges between [0, 39]
employment_status (categorical)	Employment status of the applicant. 7 possible (annonymized) values.
credit_risk_score (numeric)	Internal score of application risk. Ranges between [−191, 389].
email_is_free (binary)	Domain of application email (either free or paid).
housing_status (categorical)	Current residential status for applicant. 7 possible(annonymized) values.
phone_home_valid (binary)	Validity of provided home phone.
phone_mobile_valid (binary)	Validity of provided mobile phone
bank_months_count (numeric)	How old is previous account (if held) in months. Ranges between [−1, 32] months (-1 is a missing value).
has_other_cards (binary)	If applicant has other cards from the same banking company.
proposed_credit_limit (numeric)	Applicant's proposed credit limit. Ranges between [200, 2000].
foreign_request (binary)	If origin country of request is different from bank's country.
source (categorical)	Online source of application. Either browser (INTERNET) or app (TELEAPP)
session_length_in_minutes (numeric)	Length of user session in banking website in minutes. Ranges between [−1, 107] minutes (-1 is a missing value)
device_os (categorical)	Operative system of device that made request. Possible values are: Windows, macOS, Linux, X11, or other.
keep_alive_session (binary)	User option on session logout.
device_distinct_emails (numeric)	Number of distinct emails in banking website from the used device in last 8 weeks. Ranges between [−1, 2] emails (-1 is a missing value).
device_fraud_count (numeric)	Number of fraudulent applications with used device. Ranges between [0, 1].
month (numeric)	Month where the application was made. Ranges between [0, 7].
fraud_bool (binary)	If the application is fraudulent or not.

Table 1 : BAF Dataset Field Description

3.3. Handling Class Imbalance

To address the significant class imbalance observed in the dataset, I decided to go with the RandomUnderSampler technique, a method widely discussed in the literature on class imbalance solutions, as highlighted in (Debashree Devi et al., 2020). This approach strategically reduced the number of instances in the majority class (non-fraud accounts), thereby enhancing the prominence of the minority class (fraud accounts) and effectively mitigating the challenges associated with modeling imbalanced data.

By applying RandomUnderSampler, there was a significant transformation in the distribution of classes. The initial dominance of non-fraud accounts gave way to a more balanced representation of both fraud and non-fraud instances. This shift not only ensured that the models were not biased by the abundance of non-fraud cases but also facilitated their ability to learn patterns and features associated with fraudulent instances. This corrective measure enables to train and test the models on the dataset to provide a clearer understanding of fraudulent activities.

This process greatly enhanced the effectiveness of the machine learning endeavors, enabling the trained models to be more sensitive to the intricacies of fraudulent transactions. By reducing the class imbalance, I was able to significantly improve the models' ability to distinguish between genuine and fraudulent account openings. Consequently, by recalibrating the dataset's class distribution using the RandomUnderSampler technique, this project laid a balanced foundation for subsequent phases of model implementation and evaluation, resulting in well-informed and reliable outcomes.

```
# Define the undersampling strategy
undersample = RandomUnderSampler(sampling_strategy=0.5)

# Define the column names and target variable
cols = df.columns.tolist()
target = "fraud_bool"

# Exclude the target variable from the column list
cols = [c for c in cols if c != target]

# Define X (features) and Y (target)
X = df[cols]
Y = df[target]

# Undersample the dataset
X_under, Y_under = undersample.fit_resample(X, Y)
```

Figure 2: UnderSampler Implementation

After Undersampling, the class imbalance issue in the dataset has been addressed as is evident from the graph below.

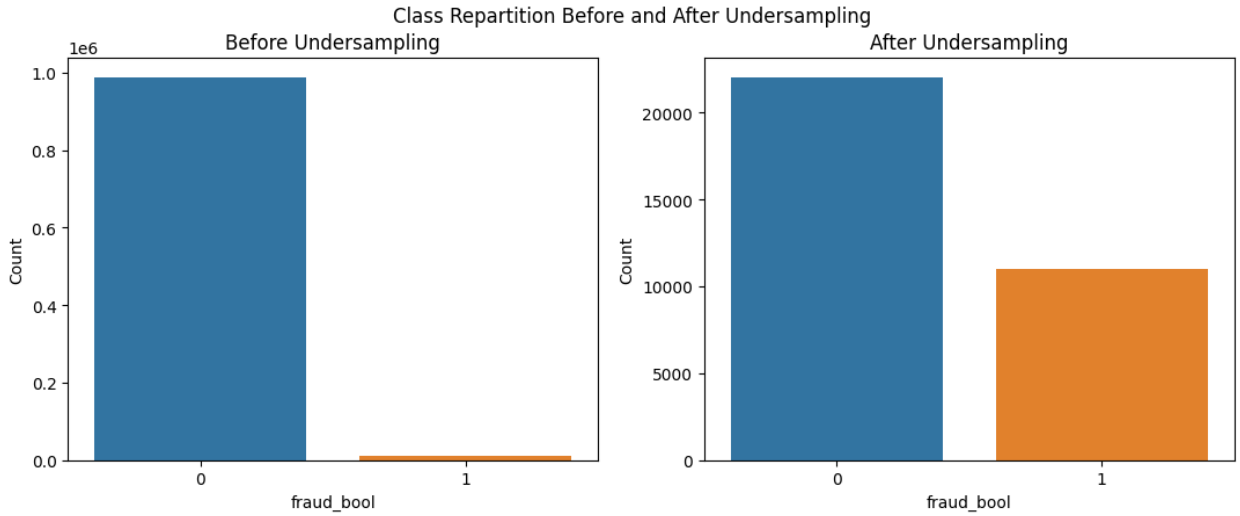

Figure 3: Class Repartition before and after Undersampling

3.4. Feature Engineering and One-hot Encoding

During the feature engineering phase, I made two important transformations to the dataset that greatly improved its compatibility with machine learning models. These transformations aimed to boost the accuracy and reliability of our fraud detection process (Chen et al. ,2020).

Firstly, I implemented a crucial preprocessing step called Standardization, which involved scaling the numerical features to have a mean of zero and a standard deviation of one. This ensured that all the numerical features were on a similar scale, which was vital in preventing any one feature from exerting undue influence on the model training process solely because of its larger magnitude. Within the realm of enhancing the dataset's compatibility with machine learning models, I took a crucial step by employing the technique of OneHotEncoding. This method serves as a vital bridge to overcome the inherent incompatibility between categorical variables and algorithms that rely on numerical input. By converting the categorical attributes into binary vectors, I prepared the data for smooth integration with the selected models, ultimately fortifying their predictive capabilities. Each unique category in the original categorical variable was transformed into a separate binary column, enabling the models to effectively analyze and discern these variables.

By implementing these thoughtfully crafted feature engineering techniques, I significantly improved the compatibility of the dataset with the machine learning models. This, in turn, led to a more accurate and dependable fraud detection process, allowing the models to precisely identify and mitigate instances of fraudulent activities.

To succinctly outline the methodology: I divided the dataset into training and testing subsets, with a clear demarcation of a 70-30 split, which facilitated the evaluation of model performance on unseen data. Subsequently, I encountered a challenge in directly incorporating the qualitative nature of categorical features into the machine learning process. That's when the OneHotEncoder surfaced as the ultimate solution. As the encodings unfolded, a discernible transformation took place. The columns that initially held categorical entries underwent an evolution, with each categorical value branching into its distinctive binary representation. The outcome was an intricately designed matrix that fluently communicated with machine learning algorithms. Moreover, the seamless fusion of numerical features with the newly engineered binary counterparts resulted in a harmonious representation of the dataset—thus setting the stage for upcoming modeling endeavors.

The significance of this transformation lies in its ability to harmonize the categorical and numerical domains, hinting at a congruence that empowers machine learning models to decipher intricate patterns across diverse attributes. As the dataset underwent a metamorphosis under the influence of One-hot Encoding, it resonated with an ensemble of models poised to extract insights and decode the complexities of fraudulent account openings. This process served as a pivotal juncture, where categorical restrictions were relinquished, and the landscape of predictive modeling expanded to navigate the intricacies of financial data with enhanced precision and accuracy.

```
# random_state: the seed used by the random number generator for reproducibility
X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.3, random_state=1)

# OneHotEncode on all the categorical features
# Find all the columns containing categorical features

s = (X_train.dtypes == 'object') # list of column-names and whether they contain categorical features
object_cols = list(s[s].index) # All the columns containing these features

# Create an instance of the OneHotEncoder
ohe = OneHotEncoder(sparse=False, handle_unknown='ignore')

# Get one-hot-encoded columns for the training data
ohe_cols_train = pd.DataFrame(ohe.fit_transform(X_train[object_cols]))

# Get one-hot-encoded columns for the test data
ohe_cols_test = pd.DataFrame(ohe.transform(X_test[object_cols]))

# Set the index of the transformed data to match the original data
ohe_cols_train.index = X_train.index
ohe_cols_test.index = X_test.index

# Remove the original categorical columns from the training and test data
num_X_train = X_train.drop(object_cols, axis=1)
num_X_test = X_test.drop(object_cols, axis=1)

# Concatenate the numerical data with the transformed categorical data for the training data
X_train = pd.concat([num_X_train, ohe_cols_train], axis=1)

# Concatenate the numerical data with the transformed categorical data for the test data
X_test = pd.concat([num_X_test, ohe_cols_test], axis=1)
```

Figure 4: One-hot Encoding implementation

3.5. Train-Test Split

During the train-test split phase, I partitioned the dataset into two distinct sets: the training set and the testing set in a 70:30 ratio, guaranteeing balanced class distributions in both sets.

This crucial step involved the utilization of the train_test_split() function from the scikit-learn library. By splitting the data in this manner, the main aim is to evaluate the performance of the four models on unseen data, thereby simulating real-world scenarios and verifying their ability to generalize. Training the models on the training set and subsequently assessing their predictions on the testing set enabled to effectively gauge their accuracy when confronted with novel and unfamiliar data. This approach not only facilitated the identification of potential overfitting problems but also ensured that the models could detect fraudulent transactions in a robust and reliable manner. I elected to train and evaluated four machine learning models - XGBoost, SVM, MLP Classifier, and Multilayer Neural Network - on the under sampled training data.

3.6. Model Selection

The selection of the four machine learning models for fraud detection in this research project is based on their proven effectiveness in similar domains and extensive usage in the literature. Each model offers unique strengths and capabilities that can contribute to a comprehensive evaluation of fraud detection performance. The justification for their selection is grounded in previous studies and their respective advantages in handling fraud detection challenges.

3.6.1. Support Vector Machine

SVM is a renowned and formidable classifier, widely recognized for its capability to handle intricate decision boundaries. Regarding fraud detection, where data often exhibits nonlinear relationships, SVM proves to be advantageous as it efficiently identifies an optimal separating hyperplane through the utilization of the kernel trick. Bhattacharyya et al. (2015) have effectively demonstrated the effectiveness of SVM specifically in credit card fraud detection scenarios. Our selection of SVM aims to complement XGBoost and provide a comprehensive comparison between these two classifiers, thereby enhancing the overall analysis and evaluation process.

It is an incredibly powerful classification algorithm with numerous advantages. Its main goal is to find the best hyperplane that separates data into different classes while maximizing the distance between those classes. What makes SVM truly remarkable is its ability to handle non-linearly separable data, thanks to the incredible kernel trick that transforms data into a higher-dimensional space.

The benefits of this model are vast. One remarkable advantage is its remarkable capability to effectively handle data with a high number of dimensions. Whether the dataset is complex or intricate, SVM can efficiently tackle it with ease. Moreover, SVM's utilization of kernel functions enables the creation of intricate decision boundaries, enabling a more insightful exploration of the data. This characteristic proves particularly advantageous when dealing with non-linear data. Furthermore, SVM demonstrates commendable generalization capabilities, making accurate predictions even on previously unseen, novel data points. This attribute serves as a testament to the reliability and robustness of SVM. Additionally, SVM consistently delivers impressive results in classification tasks. Whether it's a binary classification problem or a multiclass task at hand, SVM's methodology ensures excellent performance. Its ability to handle complex data and create precise decision boundaries contributes to its effectiveness.

The architecture involves the creation of a hyperplane that optimally separates the classes and maximizes the margin between them. The data points closest to this hyperplane play a crucial role in accurate classification. The model finds the hyperplane that maximizes the margin between the support vectors of different classes. The choice of the kernel function determines the transformation of data into a higher-dimensional space, enabling SVM to accomplish non-linear separation. By formulating a constrained optimization problem, SVM aims to locate the optimal hyperplane and subsequently classify new data points based on their position relative to the hyperplane.

3.6.2. XGBoost

XGBoost, a powerful gradient boosting algorithm, has gained widespread recognition in various domains, including fraud detection. It has shown exceptional performance in handling large datasets and achieving high levels of accuracy and precision. Previous studies have confirmed that XGBoost outperforms other traditional machine learning algorithms in fraud detection tasks (Lei et al., 2020). In this project, the aim is to utilize XGBoost for identifying fraudulent bank payments due to its robustness and excellent predictive capabilities.

XGBoost is a highly flexible and powerful gradient boosting algorithm that has gained a strong reputation for its exceptional performance. By creating an ensemble of weak learners, typically decision trees, XGBoost consistently improves their predictive capabilities by minimizing a loss function. This algorithm is particularly skilled at handling complex relationships within data. The benefits of XGBoost are manifold. Firstly, it allows for extensive customization, ensuring compatibility with various data types. Secondly, XGBoost incorporates regularization techniques that effectively combat overfitting, enhancing the reliability of its predictions. Thirdly, this algorithm excels at analyzing both structured and unstructured data, showcasing its versatility. Lastly, XGBoost demonstrates remarkable proficiency in managing missing values, further expanding its applicability.

In terms of the model architecture, XGBoost constructs a sequence of decision trees, continuously learning from the mistakes of the previous trees. Each subsequent tree is intricately designed to rectify the errors made by its predecessors. Finally, the culmination of predictions from these trees is determined through a weighted summation. During training, XGBoost efficiently calculates gradients and Hessians of the loss function, enabling the identification of optimal splits for each tree. XGBoost stands as a cutting-edge algorithm that excels in handling complex data analysis tasks. Its ability to adapt to diverse data types, mitigate overfitting, perform well on a wide range of datasets, and handle missing values solidifies its reputation as a robust and comprehensive machine learning model.

3.6.3. MLP Classifier from Scikit-learn

Artificial Neural Networks (ANNs) have shown exceptional performance in capturing complex patterns and relationships within data. One widely recognized implementation of ANNs is the MLP Classifier, available in the Scikit-learn library. ANNs have been previously proven effective in fraud detection tasks due to their ability to learn intricate features. For our research project, we have incorporated the MLP Classifier to leverage the advantages of ANNs and efficiently assess its performance compared to other models.

The MLP Classifier is a type of Artificial Neural Network that consists of interconnected nodes, known as neurons, organized in multiple layers. Each neuron processes its input and passes on the output to the subsequent layer. The MLP excels at identifying complex patterns in data and is highly suitable for tasks such as image recognition, speech recognition, and fraud detection. One notable feature of the MLP is its ability to handle both linear and non-linear relationships. Its model architecture comprises an input layer, one or more hidden layers, and an output layer, with each layer containing multiple neurons that apply activation functions to their respective inputs. A critical aspect of the MLP's functionality lies in its training process. By utilizing backpropagation and gradient descent, the network adjusts the weights and biases of its neurons to minimize the loss function. This meticulous fine-tuning process enables the model to make accurate predictions.

3.6.4. Multilayer Neural Network with TensorFlow/Keras

Although the MLP Classifier from Scikit-learn offers a convenient implementation of Artificial Neural Networks (ANNs), we have opted to develop a custom Multilayer Neural Network using TensorFlow and Keras. This decision allows us to have more flexibility and customization in our network's design. By constructing the neural network from scratch, we can meticulously tailor the architecture and hyperparameters to optimize its performance for the specific BAF dataset, which is crucial for fraud detection. This personalized approach enables us to exploit the unique characteristics of the data and potentially achieve superior results.

There are several advantages to using a custom neural network architecture. One of the main benefits is the ability to tailor the network's structure to the specific problem, unleashing its full potential for achieving optimal results. TensorFlow and Keras provide a comprehensive set of tools that allow us to construct intricate neural network architectures of varying complexities. This versatility makes our model capable of handling different types of data, such as numerical, image, or text, effectively.

When designing architecture, we have the freedom to experiment with different combinations of layers, neurons, and activation functions. This flexibility empowers us to customize the network's structure to address the specific needs of the problem, enhancing its efficiency and accuracy.

Furthermore, TensorFlow and Keras implement automatic differentiation during training, enabling efficient backpropagation through the computation of gradients. By selecting the appropriate optimizer, loss function, and evaluation metrics, we can fine-tune the model's parameters to minimize loss and maximize performance. Our custom Multilayer Neural Network using TensorFlow and Keras provides an ideal framework for designing and refining neural network architectures tailored to meet the specific demands of the problem. With its flexibility, versatility, and powerful optimization techniques, this model serves as an asset in the field of machine learning.

3.7. Model Training

For the implementation of the model, I made use of specific libraries and frameworks that were intricately tailored to each algorithm that was selected. To implement XGBoost and SVM, I opted for the Scikit-learn library, renowned for its efficiency and user-friendly application programming interfaces (APIs) for these particular algorithms. Similarly, for the MLP Classifier which represents the Artificial Neural Network, I also utilized the Scikit-learn library. To allow for greater flexibility and customization, I decided to design a custom Multilayer Neural Network using TensorFlow/Keras. These frameworks enabled me to meticulously refine the architecture and hyperparameters, ultimately ensuring that the neural network model performed at its optimum level. Through the utilization of these libraries and frameworks, my primary objective was to develop robust and accurate fraud detection models, with each model effectively harnessing its individual strengths and capabilities to effectively address and combat the unique challenges associated with identifying fraudulent transactions.

3.7.1. Evaluation Metrics

In this project, I have employed a range of metrics, including Accuracy, Precision, Recall, F1 Score, Area Under the ROC Curve (AUC-ROC), and Area Under the Precision-Recall Curve (AUC-PR), to

evaluate the performance of the models that I have selected in detecting bank account fraud using the BAF dataset.

Accuracy is a widely used metric that evaluates the percentage of correctly predicted instances out of all the predictions made. It is computed by summing the number of True Positives and True Negatives, and then dividing it by the total number of predictions. Although accuracy is a valuable measure for datasets with balanced classes, it may not accurately assess the performance of a model when dealing with imbalanced datasets. For instance, if there is a significant imbalance between the number of non-fraud instances and fraud instances, a model that predicts all instances as non-fraud can still achieve a high accuracy score. Hence, while it is important, it should be considered in conjunction with other metrics.

The precision metric assesses the accuracy of positive predictions made by the model. By dividing the number of True Positives by the sum of True Positives and False Positives, it determines the percentage of correctly identified positive instances. Precision is especially valuable in cases where false positives are undesirable, as it measures the model's ability to avoid misclassifying non-fraudulent cases as fraudulent. Higher the precision value means when the model flags a transaction as fraudulent, the chances are high that it is correct.

Recall, also referred to as sensitivity or true positive rate, measures the percentage of accurately predicted positive instances compared to the total number of actual positive instances. It is calculated by dividing the true positives by the sum of true positives and false negatives. Understanding recall is crucial in assessing the effectiveness of a model in identifying true positive cases, especially in fraud detection. Higher recall value signifies that the model excels in recognizing instances of fraud within the total instances of fraud.

The F1 Score plays a vital role in machine learning because it combines precision and recall, giving us a comprehensive view of how well a model performs. It is calculated by taking the harmonic mean of precision and recall, which helps strike a balance between false positives and false negatives. This metric proves particularly valuable when dealing with imbalanced datasets.

The confusion matrix holds immense importance when it comes to assessing the effectiveness of classification models. It serves as a concise summary of the predictions made by the model in comparison to the actual class labels. Within the matrix, we consider True Positives, False Positives, True Negatives, and False Negatives. By leveraging this powerful tool, we gain a visual representation of where our model's predictions align with or deviate from the actual labels, enabling a comprehensive assessment of its performance.

The Area Under the ROC Curve (AUC-ROC) serves as a powerful metric for evaluating model performance. It allows us to assess the balance between the True Positive Rate and the False Positive Rate across different thresholds, thereby gauging a model's ability to distinguish between classes. A higher AUC value signifies superior performance, with the ideal curve approaching the top-left corner. AUC-ROC proves particularly beneficial when comparing models and considering the trade-offs between sensitivity and specificity.

The ROC curve shows the model's capacity to distinguish between positive and negative classes visually. The trade-off between the True Positive Rate and the False Positive Rate at various categorization levels is demonstrated. The performance of the model in terms of classification accuracy improves as the curve gets closer to the top-left corner.
In addition to the ROC curve, we may also assess model performance using the Precision-Recall curve, especially in datasets with imbalances. This graph demonstrates how precision and recall trade off at various classification criteria. More accurate positive case identification while reducing false positives is indicated by a Precision-Recall curve that closely hugs the top-right corner.

3.8. Ethical Considerations

To begin with, given the study's focus on financial data and fraud detection, it is paramount to ensure the privacy and confidentiality of the utilized data. To mitigate this concern, all sensitive and personally identifiable information was anonymized and removed from the dataset prior to analysis. Moreover, the study solely relied on publicly available datasets, adhering to data usage policies, and guaranteeing compliance with pertinent data protection regulations.

Regarding the BAF dataset, stringent measures are implemented during its generation process to uphold privacy. The dataset is created utilizing cutting-edge Generative Adversarial Network (GAN) models, which introduce a level of differential privacy (Jesus et al., 2022). This is accomplished by introducing noise to the instances of the original dataset, rendering it difficult to infer specific information about individuals from the data. Additionally, personal information columns are categorized prior to training the GAN model, further safeguarding the privacy of individuals involved. By employing these privacy-preserving techniques, the BAF dataset maintains a higher level of anonymity and confidentiality, ensuring that sensitive and personally identifiable information is effectively safeguarded.

The adoption of these privacy-preserving measures aligns with ethical considerations regarding data privacy and protection. As a result, researchers can effectively carry out their analyses without compromising the privacy of individuals associated with the data. This substantiates the usability of the dataset for fraud detection research and experimentation, while adhering to ethical guidelines and regulations pertaining to data privacy and confidentiality.

Another crucial ethical consideration pertains to the potential bias that may be inherent within the data, thereby potentially influencing the outcomes of the model in a biased manner. To alleviate this concern, diligent efforts were undertaken to address this issue through meticulous preprocessing techniques, effective feature engineering, and comprehensive model evaluation. It is of utmost importance to remain vigilant regarding the potential impact of biases on the results.

3.9. Risks & Limitations

The dataset utilized in the study may not fully encompass the breadth and intricacy of real-world fraud scenarios. The accuracy and performance of the models greatly rely on the quality and representativeness of the data employed. Moreover, downsampling the data to address class imbalance poses the risk of losing information and diminishing model generalization. Machine learning models also have inherent limitations, including overfitting and incapability to handle certain data patterns. Despite meticulous tuning, some models might still struggle to generalize effectively on unseen data or perform optimally for specific types of fraud cases. Furthermore, the study primarily focused on a specific set of machine learning algorithms and did not explore other advanced techniques such as deep learning or ensemble methods. The choice of models can impact the results, and there may exist more suitable algorithms that were not taken into consideration in this study.

Data privacy and confidentiality were diligently addressed through the meticulous anonymization and removal of personally identifiable information within the dataset, thereby safeguarding individuals' sensitive data. Rigorous efforts have been made to proactively address model bias and fairness concerns through comprehensive preprocessing and evaluation, while remaining mindful of potential biases inherent in the model outcomes. However, it is essential to acknowledge that the study's generalizability may be somewhat constrained due to the utilization of publicly available datasets, potentially resulting in variation with regards to model performance when applied in real-world scenarios. To tackle the issue of data imbalance, under sampling techniques were judiciously applied, prioritizing the rectification of critical information loss. Moreover, it is imperative to acknowledge the potential risks of model overfitting, as well as the inherent challenges in interpreting complex machine learning models, particularly neural networks. To address these risks, scrupulous data handling and model evaluation were consistently employed, while ensuring utmost transparency in reporting findings and limitations. Furthermore, the study's utilization of a synthetic dataset for fraud detection, while highly informative, may not comprehensively capture the intricate complexities inherent in real-world scenarios. It was imperative to place emphasis on meticulous attention to detail, encompassing data handling procedures, comprehensive model evaluation, and the transparent reporting of findings and limitations.

3.10. Reflection

Throughout the course of this research endeavor and the exploration of various methodologies, I have gained significant insights into the realm of fraud detection and machine learning. The immersive experience of implementing different models, including XGBoost, SVM, MLP Classifier, and a custom Multilayer Neural Network, has enriched my understanding of their capabilities and limitations when faced with real-world challenges.

Undoubtedly, one of the most crucial lessons learned pertained to the criticality of data preprocessing. The manipulation of missing values, addressing data imbalance, and feature engineering are pivotal steps in ensuring the reliability and quality of the models' results. Moreover, the ethical considerations associated with working with financial data have become increasingly apparent. The utmost priority must always be accorded to privacy and confidentiality, while also taking great care to mitigate any potential biases in the data that may lead to skewed outcomes.

The decision to explore the technique of undersampling to address data imbalance yielded invaluable insights into the trade-offs between model performance and real-world applicability. Although the training process without undersampling closely mirrored actual conditions, it necessitated extended computation times and resulted in biased outcomes. Consequently, this experience underscored the significance of thoughtfully striking a balance between model performance and practicality in the field of fraud detection.

Furthermore, delving into the BAF dataset generated from a cutting-edge Generative Adversarial Network (GAN) model proved to be an intriguing endeavor. Comprehending the intricacies of how noise was injected into instances and personal information was categorized to ensure differential privacy ignited a deep curiosity within me regarding advanced data generation techniques and their role in preserving the sanctity of data privacy.

All in all, this section outlining the chosen methodology has not only honed my technical skillset but has also instilled within me a profound sense of responsibility and ethical consciousness when it comes to working with sensitive financial data. It has significantly influenced my research approach by emphasizing the paramount importance of thorough data preprocessing, impartial evaluation, and a diligent consideration of the wider implications of our findings. As I forge ahead with this research endeavor, these profound insights are certain to shape my interactions with the data and inspire me to adopt a principled and meticulous approach to untangling the complexities of fraud detection in the ever-evolving digital landscape.

Chapter 4: Analysis & Visualizations

4.1. Data Overview

The dataset utilized in this project constitutes a pivotal component in the analysis of fraud detection. It encompasses a significant volume of transactions, along with a diverse range of features that possess the potential to aid in the identification of fraudulent activities. This dataset comprises both numerical and categorical features, with the target variable denoted as "fraud_bool," which signifies the occurrence of fraudulent or non-fraudulent transactions.

Understanding of the distribution of the target variable is important. It is highly likely that the dataset exhibits an imbalance, given that instances of fraudulent transactions are comparatively rare in relation to legitimate ones. This imbalance can pose challenges during the training of models, thereby impacting their efficacy in accurately discerning cases of fraud.

The numerical features within the dataset provide valuable insights into the characteristics of transactions, such as income, customer age, intended transaction amount, and credit risk score. Scrutinizing the summary statistics pertaining to these features aids in the identification of potential outliers and facilitates an assessment of their influence on the overall data distribution.

As for the categorical features, encompassing payment type, employment status, housing status, source, and device operating system, they assume a critical role in comprehending the distinct transaction profiles. Analyzing the frequency distribution of these categorical features serves to pinpoint the most prevalent transaction scenarios and ascertain any discernible patterns that differentiate fraudulent transactions from legitimate ones.

Furthermore, when dealing with categorical features, I employed the powerful technique of one-hot encoding. By transforming categorical variables into a binary format, this methodology empowers our machine learning algorithms to effectively leverage the information contained within these features. Consequently, our models can intelligently account for the distinct attributes represented by the various categorical variables, further enriching the insights gleaned from the data.

Additionally, I addressed the issue of feature scaling for the numerical attributes. Through the process of feature scaling, I standardized the range of these variables, ensuring that they do not unfairly dominate the learning process solely due to their scale. This normalization technique provides a level playing field for all features, enabling a more equitable contribution from each attribute, and facilitating a more accurate analysis and model interpretation.

4.2. Descriptive Statistics & Exploratory Data Analysis (EDA)

To develop a comprehensive understanding of the dataset, I undertook an extensive examination using descriptive statistics and exploratory data analysis (EDA). This meticulous process involved analyzing various characteristics of the data, including measures of central tendencies, distributions, and the interrelationships between variables. Through the implementation of these rigorous analyses, my objective was to extract meaningful insights and identify discernible patterns that could furnish valuable information for the discernment of fraudulent activity.

	count	mean	std
fraud_bool	1000000.0	0.011029	0.104438
income	1000000.0	0.562696	0.290343
name_email_similarity	1000000.0	0.493694	0.289125
prev_address_months_count	1000000.0	16.718568	44.046230
current_address_months_count	1000000.0	86.587867	88.406599
customer_age	1000000.0	33.689080	12.025799
days_since_request	1000000.0	1.025705	5.381835
intended_balcon_amount	1000000.0	8.661499	20.236155
zip_count_4w	1000000.0	1572.692049	1005.374565
velocity_6h	1000000.0	5665.296605	3009.380665
velocity_24h	1000000.0	4769.781965	1479.212612
velocity_4w	1000000.0	4856.324016	919.843934
bank_branch_count_8w	1000000.0	184.361849	459.625329
date_of_birth_distinct_emails_4w	1000000.0	9.503544	5.033792
credit_risk_score	1000000.0	130.989595	69.681812
email_is_free	1000000.0	0.529886	0.499106
phone_home_valid	1000000.0	0.417077	0.493076
phone_mobile_valid	1000000.0	0.889676	0.313293
bank_months_count	1000000.0	10.839303	12.116875
has_other_cards	1000000.0	0.222988	0.416251
proposed_credit_limit	1000000.0	515.851010	487.559902
foreign_request	1000000.0	0.025242	0.156859
session_length_in_minutes	1000000.0	7.544940	8.033106
keep_alive_session	1000000.0	0.576947	0.494044
...			
session_length_in_minutes	8.866131	85.899143	
keep_alive_session	1.000000	1.000000	
device_distinct_emails_8w	1.000000	2.000000	
month	5.000000	7.000000	

payment_type	
AB	370554
AA	258249
AC	252071
AD	118837
AE	289

	count	unique	top	freq
payment_type	1000000	5	AB	370554
employment_status	1000000	7	CA	730252
housing_status	1000000	7	BC	372143
source	1000000	2	INTERNET	992952
device_os	1000000	5	other	342728

	fraud_bool	count
0	0	988971
1	1	11029

Figure 5: Descriptive Statistics

In dissecting the BAF dataset, we gained valuable insights into its key attributes through descriptive statistics. The column labeled "fraud_bool" shed light on the distribution of classes, revealing that most transactions (98.9%) were deemed non-fraudulent, while a smaller portion (1.1%) were identified as fraudulent. Summary statistics unveiled intriguing patterns, such as an average income of approximately 0.56 with some notable variation, and an average customer age of around 33.7, accompanied by a diverse range spanning a standard deviation of 12.03. Certain columns, namely "velocity_6h," "velocity_24h," and "velocity_4w," displayed higher mean values, indicating frequent transactional activities. Furthermore, the average credit risk score hovered around 131. Notably, the dataset's imbalanced class distribution emphasizes the necessity of utilizing specialized techniques in model training, to counteract the potential impact of skewed data on predictive performance. Armed with these valuable insights, our subsequent analyses will encompass feature exploration, correlation assessment, and the development of

machine learning models, all of which are vital components in achieving our fraud detection objectives.

I conducted the initial exploratory data analysis (EDA) in Tableau, where I developed an interactive dashboard. This dashboard features two tabs: one for fraud and another for non-fraud data, facilitating the separation and analysis of these categories. Users can easily switch between these tabs using a filter, allowing for a dynamic exploration of the dataset. I have attched a snippet of the dashboard below. The dashboard has been deployed to Tableau Public and the link for the same is provided in the Appendix.

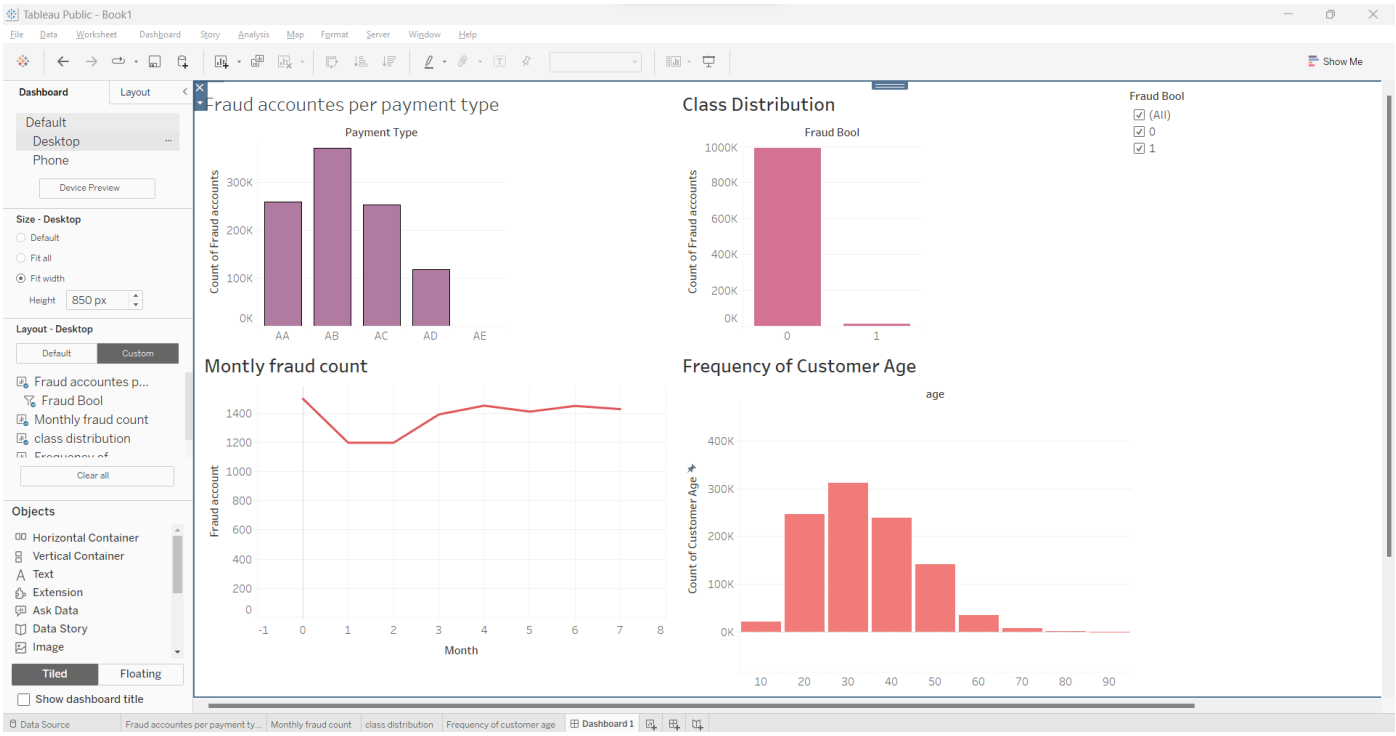

Figure 6: EDA Dashboard snippet from Tableau

Number of Fraudulent Accounts by Payment Type: The bar graph illustrates the distribution of fraud accounts across different payment methods, revealing a fascinating pattern. Amongst all payment types, payment type AB stands out as the most frequently utilized method for fraudulent activities. This valuable insight indicates that perpetrators may be exploiting vulnerabilities in payment type AB, warranting a closer examination of this payment method to bolster security measures and improve fraud detection protocols.

The bar graph depicting the distribution of classes showcases a notable imbalance, with a considerable number of non-fraudulent accounts in comparison to fraudulent accounts. This disparity could potentially present challenges during the training of the model, as algorithms may exhibit bias towards predicting the majority class. Consequently, it becomes imperative to

address this imbalance to ensure effective training of the model for the detection of fraud cases, which represent the minority class.

Monthly Fraud Count Line Graph: The dynamic nature of the line chart illustrating the monthly fraud count implies a fluctuating pattern of fraudulent account openings across different months. This observation emphasizes the significance of implementing dynamic fraud detection strategies that can adapt to evolving patterns. Seasonal variations, economic factors, or even specific events may influence the frequency of fraudulent activities, underscoring the necessity to deploy models capable of capturing these nuanced trends.

The analysis of the frequency distribution of customer age reveals an interesting pattern. It is evident that individuals in the age range of 30-40 have the highest number of occurrences of fraudulent account openings. This finding suggests that this age group might be more vulnerable to fraud or potentially targeted by fraudsters. Furthermore, a gradual decline in fraudulent account openings is observed in the subsequent age groups. Interestingly, individuals aged 80 and above exhibit the lowest occurrence of fraudulent account openings. This trend could be attributed to the cautious behavior displayed by older individuals regarding online account activities.

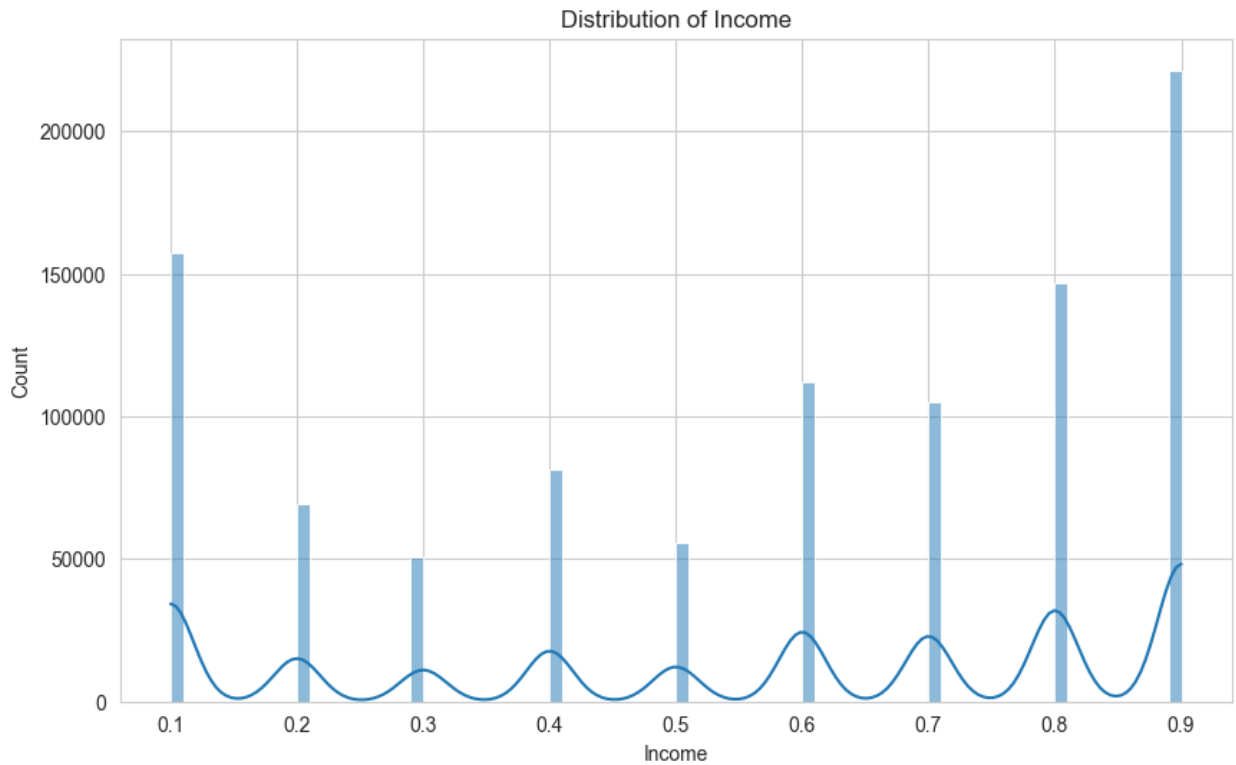

Figure 7: Distribution of Income

The histogram of income distribution indicates that the dataset is skewed towards certain income levels. Specifically, there is a significant concentration of data instances at income levels of 0.1 and 0.8, suggesting that these income categories are more prevalent in the dataset. Conversely, income levels such as 0.2, 0.3, and 0.5 have fewer instances, indicating that these income categories are less represented. This skewed distribution of income may have implications for fraud detection, as it suggests that certain income categories might be more susceptible to fraudulent activities than others. Further analysis is needed to explore potential correlations between income levels and fraudulent transactions, which can provide valuable insights for fraud detection models.

Figure 8: Distribution of Income for Fraud vs Non- Fraud

The boxplot comparing income distribution between fraudulent (1) and non-fraudulent (0) transactions reveals interesting insights. It illustrates that fraudulent transactions tend to have slightly higher income values, as indicated by the higher median in the "fraud" category compared to "non-fraud." However, both categories exhibit a wide range of income values, with significant overlaps, implying that income alone may not be a strong predictor for identifying fraudulent transactions. This suggests that other factors or features may play a more critical role in distinguishing between fraudulent and non-fraudulent activities in the dataset. Further analysis is needed to explore these factors and their relationships with fraud detection.

Chapter 5: Model Evaluation and Discussion

5.1. Support Vector machines (SVM)

In my journey through model training, I embarked on a comprehensive exploration to uncover the intricate dynamics of the Support Vector Machine (SVM). By implementing an instance of the SVC model, the training phase unfolded, marking a pivotal moment in my pursuit of predictive mastery. The subsequent predictions on the test data yielded a wealth of crucial evaluation metrics that formed the foundation for profound insights into the performance of the model.

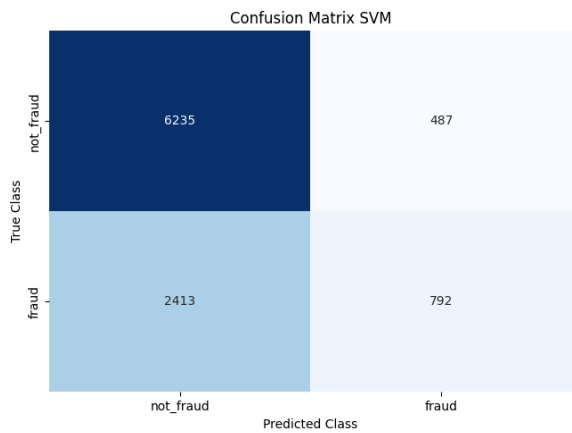

Figure 9: SVM - Confusion matrix

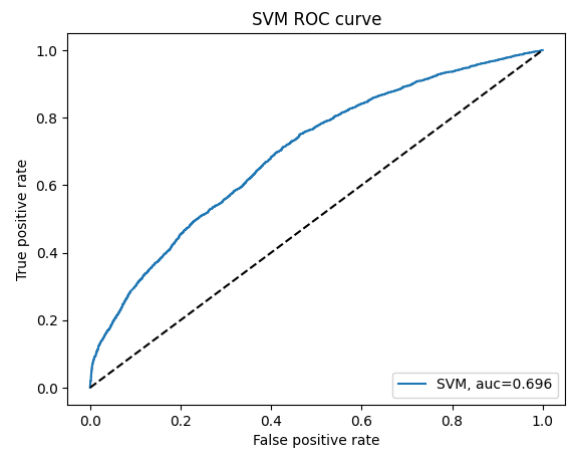

Figure 10: SVM - ROC curve

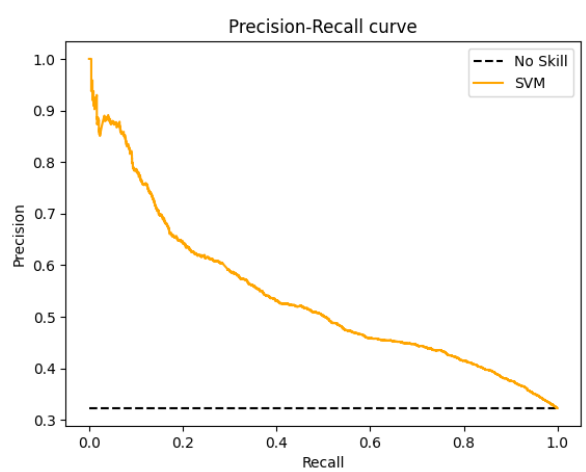

Figure 11: SVM - Precision-recall curve

Several indicators were employed to evaluate the models. Accuracy gauged the overall correctness of the model's predictions. Precision and Recall measured the model's ability to correctly classify positive instances and identify actual positive instances, respectively. The F1 Score, blending precision and recall harmoniously, provided a holistic view of the model's

effectiveness. The derived accuracy of 70.8% exemplified the model's propensity to make accurate predictions, while a precision of 61.9% underscored its capability to pinpoint true positive cases. However, a recall of 24.7% unveiled the model's limitations in capturing all positive instances. The intricacies of these metrics merged to form a mosaic that depicted SVM's performance in the realm of fraud detection.

Furthermore, the Confusion Matrix measured model's classification acumen. The juxtaposition of actual and predicted classes offered valuable insights. Values of 6235 true negatives and 792 true positives symbolized accurate classifications, while 487 false positives and 2413 false negatives denoted instances where predictions fell short. This matrix, resembling a blueprint, provided a granular view of the model's classification prowess.

Delving deeper, the ROC gauged SVM's discrimination ability. This curve vividly portrayed the true positive rate against the false positive rate, with an AUC score of 0.6961 encapsulating the area under the curve. The AUC score lent empirical credibility to the model's aptitude in distinguishing between fraudulent and non-fraudulent accounts.

In the precision-recall domain, the curve represents a trade-off between precision and recall. The presence of the "No Skill" line, characterized by a linear progression above 0.3, provided a benchmark for the model's performance. In contrast, SVM's trajectory started at a precision recall of 1, highlighting the model's commendable precision when handling positive instances. Gradually descending, it painted an intriguing picture of the model's intricate dance between precision and recall, uncovering the delicate balance between predictive accuracy and comprehensive coverage.

Ultimately, this exploration yielded more than just metrics; it laid bare SVM's potential and limitations in the realm of fraud detection. These insights underscored the intricate interplay between algorithmic prowess and real-world intricacies, thereby paving the way for a comprehensive evaluation and a nuanced understanding of SVM's role in the broader landscape of our research.

5.2. XGBoost

I then continued the analysis employing XGBoost method, starting with the creation of the XGBClassifier instance. As I made predictions on the test data, the beacon of evaluation metrics illuminated my path, guiding me through the nuanced performance landscape of XGBoost.

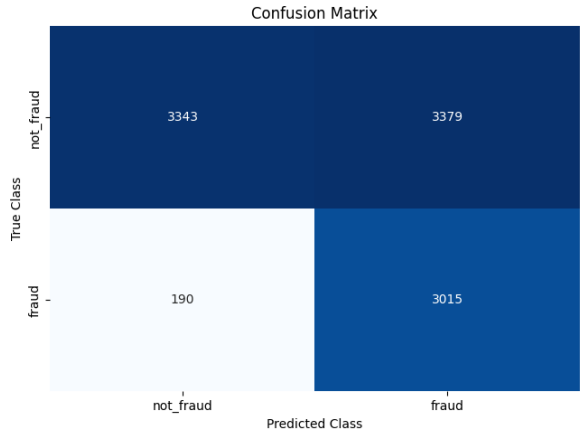

Figure 12: XGBoost - Confusion Matrix

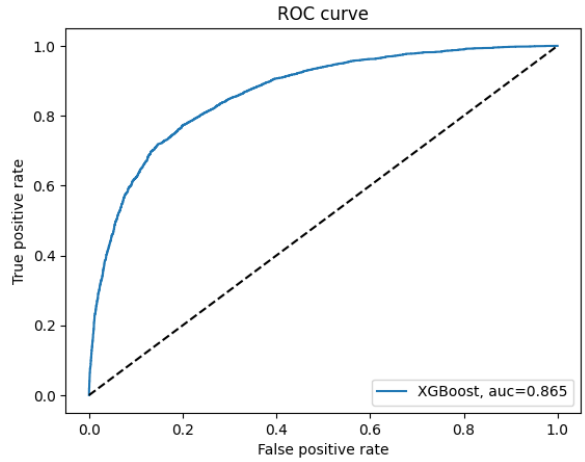

Figure 13: XGBoost - ROC curve

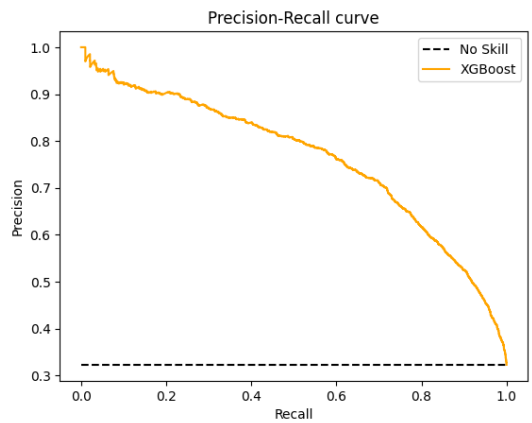

Figure 14: XGBoost - Precision-Recall curve

During this analysis, several essential metrics were utilized to assess the model's performance. Accuracy, which measures overall correctness, indicated that the model's predictions were correct 64.0% of the time. Precision and recall, two crucial metrics, played vital roles. Precision demonstrated the model's accuracy in identifying positive cases, achieving a rate of 47.2%. In contrast, recall highlighted the model's ability to capture true positives, reaching an impressive 94.1%. The F1 Score, which balances precision and recall, provided a comprehensive view of the model's performance. In essence, the analysis revealed the model's capability for accurate predictions and effective identification of actual positive cases, thereby reducing false negatives in the context of fraud detection.

Interestingly, the Confusion Matrix emerged as a visual testament, laying bare the model's classification prowess. The interplay between actual and predicted classes unfolded as a vivid reflection of the model's ability to classify accurately. The canvas painted values of 3343 true negatives and 3015 true positives, eloquently depicting precise classifications. Conversely, 3379 false positives and 190 false negatives shed light on instances where predictions deviated from

reality. This intricate tableau conveyed profound insights into XGBoost's precision in categorization.

In the realm of data visualization, the ROC curve emerged as a significant highlight, meticulously showcasing the model's discriminative capabilities. This curve, an eloquent portrayal of the true positive rate against the false positive rate, culminated in an AUC score of 0.8651, encapsulating the area under the curve. The AUC score resonated as a testament to the model's proficiency in differentiating between fraudulent and non-fraudulent accounts.

The Precision-Recall curve introduced a distinct perspective, elucidating the intricate interplay between precision and recall. In this intricate choreography, the "No Skill" line provided a crucial reference point for evaluating the model's performance. The trajectory of XGBoost, characterized by an initial precision-recall of 1, highlighted the model's precision in handling positive instances. As it meandered through its course, the curve elegantly converged with the "No Skill" line, symbolizing a finely tuned balance between predictive precision and comprehensive coverage.

This expedition into the realm of XGBoost unearthed insights that transcend mere numbers and visualizations. These insights demystified the model's capabilities, emphasizing its potential and limitations in the intricate domain of fraud detection. Ultimately, this exploration strengthened my understanding, offering a panoramic view of XGBoost's role within the broader landscape of my research endeavors.

5.3. Multi Layer Perceptron (MLP)

I started exploring the Multilayer Perceptron (MLP), a type of neural network with hidden layers and complex calculations. I used a tool called MLPClassifier to train this neural network. During training, I found that using a solver called 'lbfgs' and setting the hidden layer sizes to (100, 100) worked well. This gave me insight into how this neural network is structured and operates.

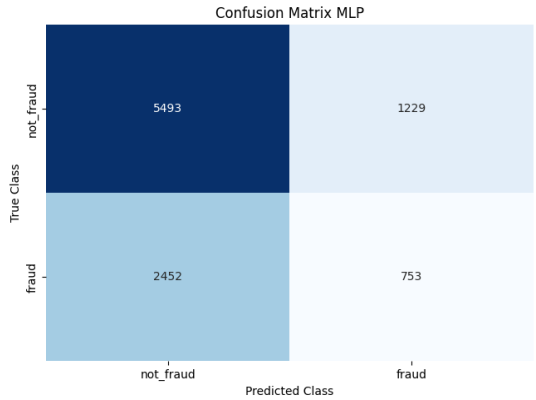

Figure 15: Confusion Matrix MLP

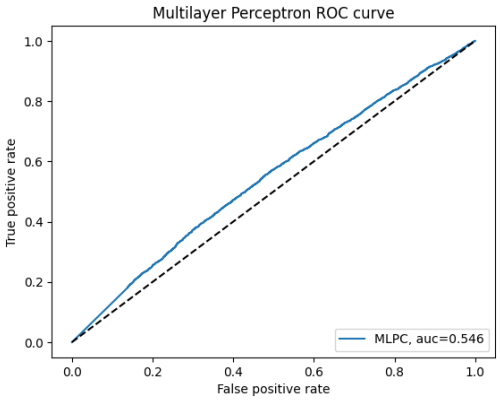

Figure 16: ROC curve MLP

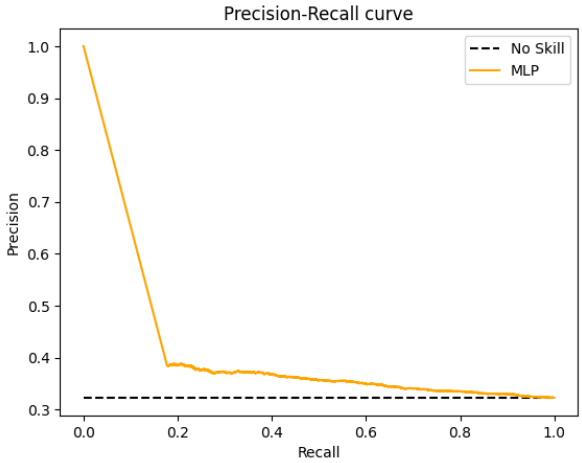

Figure 17: Precision-Recall curve MLP

As I navigated the landscape of data, the model's predictions showcased its remarkable predictive abilities. Evaluation metrics served as my compass, offering invaluable insights into the capabilities of MLP. The accuracy metric stood as a guardian of correctness, capturing the alignment between the model's predictions and reality. Precision and recall emerged as guiding beacons, each unveiling a unique aspect of the model's behavior. Precision reflected the model's accuracy in positive predictions, while recall unveiled its proficiency in capturing true positive instances. The artful blend of precision and recall, known as the F1 Score, provided a comprehensive picture of MLP's ability to strike a balance.

Throughout the evaluation journey, the model achieved an accuracy of 62.9%, reflecting its capability for accurate predictions. A precision of 37.9% highlighted its precision in predicting true positive cases. With a recall of 23.5%, MLP demonstrated its proficiency in capturing actual positive instances. The F1 Score, showcasing the harmonic union of precision and recall, amounted to 29.0%, portraying the overall effectiveness of the model.

The Confusion Matrix, a visual testament, vividly portrayed the model's classification prowess. It unveiled true negatives (5493) and true positives (753), accentuating its accuracy in classifications. On the flip side, false positives (1229) and false negatives (2452) provided insight into instances where predictions diverged from reality, contributing to a nuanced narrative of MLP's classification capabilities.

The ROC curve, illustrated the model's ability to discern between fraudulent and non-fraudulent accounts. Its culmination in an AUC score of 0.5463 elegantly encapsulated its power of discrimination, showcasing its ability to differentiate between positive and negative instances.

The Precision-Recall curve, showcased the model's delicate dance between precision and recall. Against the backdrop of the "No Skill" line, the curve embarked on a journey marked by high

initial precision, followed by a gradual decline. Ultimately, the curve gracefully converged with the "No Skill" line, illustrating the precision-recall equilibrium achieved by the model.

Hence, delving into the intricacies of MLP offered a deeper understanding that extended beyond numerical data and visual representations. It allowed to grasp the nuances of how this neural architecture functions in the context of fraud detection. This reflection significantly enriched my insights into the capabilities and limitations of MLP within the scope of this research.

5.4. Multilayer Neural Network with Tensorflow/Keras

In my exploration of Multilayer Neural Networks using TensorFlow and Keras, my aim was to uncover intricate patterns within the data. To start, I made adjustments to the data format, ensuring it was suitable for the model's requirements.

I constructed the model gradually, akin to assembling building blocks. This type of model is akin to arranging various components in a sequence. Each block was meticulously crafted, initially incorporating adjustments to enhance the model's performance. Subsequently, denser blocks were introduced, each possessing unique features to capture diverse aspects of the data. To prevent the model from fixating on specific details, I incorporated elements that helped maintain a balanced and less detailed approach.

To gauge the model's effectiveness, I utilized specific measures. Accuracy provided an indication of the model's correct predictions, while the AUC score evaluated its ability to distinguish between fraudulent and non-fraudulent cases. Additionally, metrics such as F1 Score, Precision, and Recall furnished more comprehensive insights into the model's performance.

Throughout the learning process, I guided the model's development utilizing a specific methodology, adjusting the learning rate as necessary. If the model appeared to be absorbing an excessive amount of information from the data, I implemented a stop mechanism to prevent overfitting. Thus, the model was fine-tuned to strike a balance between extracting valuable information and avoiding excessive reliance on the data.

With each round of training, the model yielded intriguing discoveries. For instance, during the last round, the model's loss stood at 0.45, indicating gradual improvement. Furthermore, its accuracy hovered around 79.3%, underscoring its proficiency in making predictions. The AUC score of 0.8014 suggested its effectiveness in discerning fraudulent from non-fraudulent cases. Detailed metrics like F1 Score, Precision, and Recall contributed additional layers of comprehension, revealing the model's adeptness in handling various prediction aspects.

The culmination of my study's findings is based on analyzing four employed models: SVM, XGBoost, MLP, and Multilayer Neural Network with TensorFlow/Keras. When closely examining these models, it becomes evident that the Multilayer Neural Network with TensorFlow/Keras achieves the highest accuracy of 79.3%. This demonstrates the immense potential of neural networks in capturing intricate patterns within financial data and effectively identifying

fraudulent activities. While all models displayed varying degrees of predictive capabilities, the superiority of the Multilayer Neural Network in terms of accuracy underscores its significance as a powerful tool for combating fraud in the financial domain.

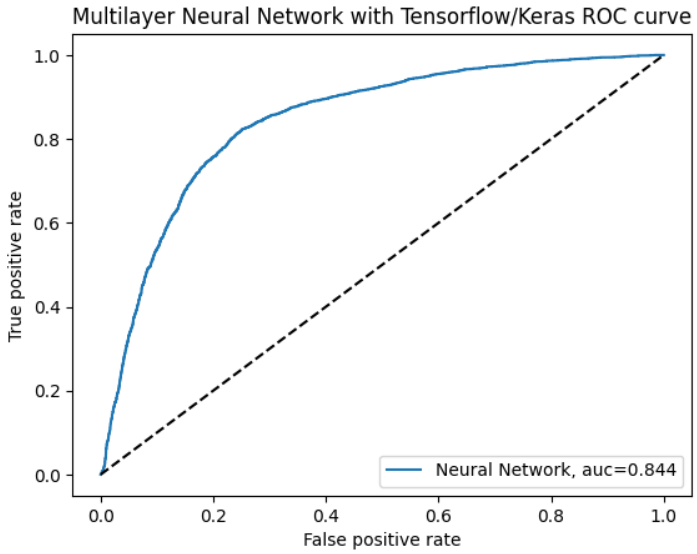

Figure 18: Multilayer Neural Network with Tensorflow/Keras - ROC curve

The ROC curve above showcases the performance of the Multilayer Neural Network with Tensorflow/Keras, with an AUC of approximately 0.927. This indicates the model's ability to effectively discriminate between positive and negative instances.

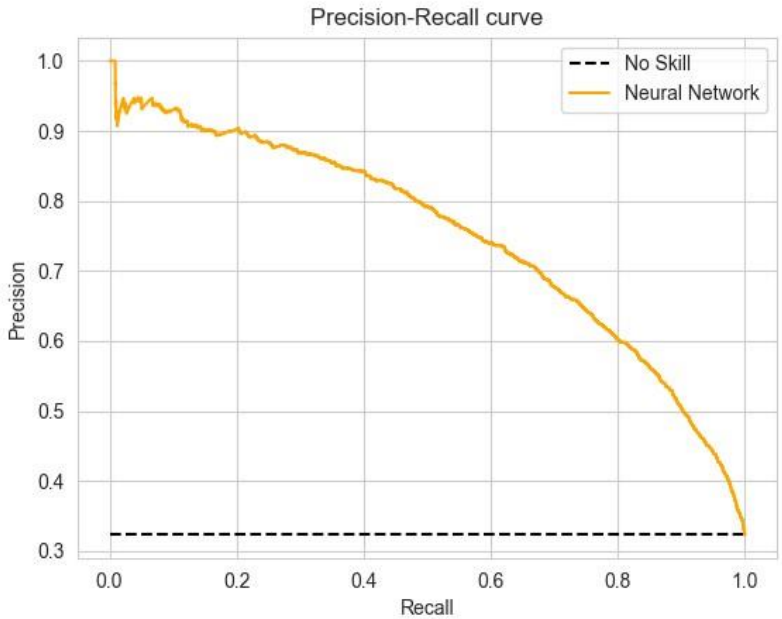

Figure 19: Neural Network with Tensorflow/Keras - Precision-Recall curve Multilayer

The Precision-Recall curve indicates that at the lowest recall values, the model achieves a very high precision, meaning it makes highly accurate positive predictions. As recall gradually increases from there, precision follows a curve, peaking and then slowly declining as recall reaches around 0.2, indicating that the model identifies a substantial portion of the actual positive instances without making too many false positive predictions. This suggests that the model effectively balances precision and recall, a valuable characteristic for tasks like fraud detection.

Model	Accuracy
SVM	70.7%
XGBoost	64%
MLP	62.9%
Multilayer Neural Network with Tensorflow/Keras	79.3%

Table 2: Accuracy comparison of 4 models

Chapter 6 - Conclusion

Upon reflection, I have to say that this research project has been an incredibly dynamic exploration into the fascinating realm of fraud account detection. As I look back on the initial aims, objectives, and broader aspirations that were set out for this study, I am reminded of the evolving journey that has truly granted me a comprehensive understanding of the problem at hand.

Addressing the fundamental challenge of combating fraud within financial transactions has served as the captivating foundation that drove my inquiry. Working closely with my academic mentors and immersing myself in the practical context of this project, I embarked on a multidimensional path. This journey has been characterized by iterative cycles of problem framing and conceptualization. While the initial problem statement encapsulated the essence of the challenge, the progression of this project necessitated adaptations and refinements to more precisely align our focus.

The methods utilized throughout this project not only demonstrated their efficacy but also provided profound insights into their practical applications. As I systematically traversed the stages of data collection, preprocessing, feature engineering, and model selection, a robust framework emerged. However, it is important to acknowledge the stark reality of dealing with a highly imbalanced dataset, which posed a formidable hurdle. To effectively address this issue, the implementation of undersampling techniques became a vital strategy, ultimately leading to a training dataset that is better balanced.

The analytical phase witnessed the convergence of model development and evaluation. The SVM, XGBoost, MLP, and Multilayer Neural Network models were each carefully tailored to address the intricacies of fraud account detection. The diverse performances of these models underscored the complexity of the task at hand. Extracting invaluable insights from the outcomes of each model served to furnish a comprehensive overview, shedding light on their respective strengths and limitations.

Beyond the objective evaluation of model performance metrics, the implications of this study extended deeply into the realm of ethics. Ethical considerations permeated every aspect of this project, emphasizing the critical need for responsible data management and unwavering vigilance against inherent biases. The delicate balance between accuracy and fairness emerged as a central focal point, influencing not only the analysis itself but also sparking impassioned ethical discussions surrounding the application of AI-driven financial fraud detection.

In summation, I can confidently say that this research expedition has been an immersive and illuminating endeavor, casting a bright light upon the intricate landscape of fraud account detection. The journey that has taken me from formulating the problem to implementing methods, conducting rigorous analysis, and grappling with ethical considerations has underscored the nuanced interplay between theory and application. Through this project, I have deepened my comprehension of both data science and its practical implications in addressing real-world challenges, thereby affirming the inherent synergy that exists between academia and industry.

Chapter 7 - Recommendations

Findings from this project will be valuable for researchers, potential collaborators from the financial industry and regulatory bodies.

In the realm of fraud detection and the use of machine learning models, it is essential to prioritize a robust ethical framework. Aligning the deployment of each model with principles of privacy, fairness, and transparency is crucial for ensuring accuracy while upholding ethical considerations. It is also important to encourage collaborations that bring together the knowledge and expertise of data scientists, industry specialists, and regulatory authorities. Such collaborations can strengthen the practicality and relevance of the developed models, enhancing their effectiveness and ethical foundations.
Embracing an iterative approach to problem framing is vital, as it allows for adaptability in addressing evolving fraud patterns and technological advancements. To enhance model accuracy, addressing class imbalance within datasets is pivotal. Undertaking measures such as undersampling ensures equal representation of both fraudulent and non-fraudulent instances, fostering unbiased model outcomes.

Prioritizing models that offer transparency and interpretability is crucial. Gaining a deep understanding of these models' inner workings enables the deciphering of their decision-making processes, fostering trust and accountability. Regular monitoring and evaluation mechanisms should be incorporated for deployed models to highlight potential biases or inaccuracies, enabling timely adjustments to maintain both effectiveness and ethical soundness.

It is also important to foster meaningful collaborations between financial entities and researchers, encouraging the exchange of anonymized data and insights to develop robust and operationally relevant models. Initiating dialogues with regulatory bodies to promote comprehensive ethical discussions surrounding AI-driven fraud detection is crucial, collaborating to establish standardized guidelines that uphold ethical AI practices within the financial domain.

In conclusion, this research advocates for an ethical, collaborative, and adaptable approach in the domain of machine-aided fraud detection. It urges stakeholders to navigate the AI landscape with a conscientious spirit, valuing precision alongside ethical integrity. By following this trajectory, we can pave the way for a financial arena marked by predictive efficacy, fairness, and ethical awareness.

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Appendix 1: Project Proposal

Online Payments Fraud Detection

Aim and Objectives

This research aims to compare the performance of different machine learning algorithms for online payments fraud detection using real-world datasets.

- Prepare and preprocess the Bank Account Fraud dataset.
- Select machine learning algorithms: SVM, XGBoost, MLP classifier, and Multilayer Neural Network.
- Identify relevant features for fraud detection.
- Train the models and optimize hyperparameters.
- Evaluate model performance using ROC curve, TPR, and FPR.
- Conduct a comprehensive comparative analysis to determine the best-performing algorithm.
- Assess model interpretability and generalization.
- Provide recommendations for the most suitable algorithm for online payments fraud detection.

Problem Statement

The primary objective of this research project is to compare the performance of two supervised learning models, XGBoost and Support Vector Machines (SVM), and two Artificial Neural Network (ANN) algorithms, namely the MLP classifier from the Sklearn library and a Multilayer Neural Network with TensorFlow/Keras. The comparison will be conducted on the task of identifying fraud transactions using the BAF dataset published in NeurIPS 2022. The goal is to determine which algorithm exhibits superior performance in terms of the ROC curve, True Positive Rate (TPR), and False Positive Rate (FPR) for fraud detection.

Literature Review

Fraud detection in the banking and financial sectors is a critical challenge, leading to extensive research on machine learning algorithms for this purpose. XGBoost, Support Vector Machines (SVM), and Artificial Neural Networks (ANNs) have shown promise in fraud detection. XGBoost has been successful in various domains, including e-commerce fraud detection (Lei et al., 2020).

SVMs are widely used due to their ability to handle complex decision boundaries and achieve high accuracy in credit card fraud detection (Bhattacharyya et al., 2018). ANNs capture complex patterns, outperforming traditional classifiers in credit card fraud detection.

"Fraud Detection on Bank Payments Using Machine Learning" by Pinku Ranjan et al. presents measures to fully automate fraud detection in banks. The study emphasizes the rising significance of fraud and the damages it causes to banks. Short-term processing challenges in transactions pose unique difficulties for fraud exposure. The paper conducts a feasibility study of various fraud detection methods using machine learning models, defining dataset attributes, metrics, and techniques to handle unbalanced datasets. Understanding underlying patterns, such as evolving purchasing habits of cardholders and changing tactics of fraudsters, is crucial for effective fraud detection. The authors also highlight several methods to obtain sequential features from credit card transactions.

Comparative studies have compared various algorithms, revealing that XGBoost and SVM achieve higher accuracy than others .Ensemble approaches have been effective in improving fraud detection models' accuracy and robustness. Feature engineering's role is crucial, as demonstrated by Chen et al. (2020), who compared different techniques and their impact on fraud detection models. Additionally, the "BankSim" simulator generates synthetic data for fraud detection research, ensuring privacy and ease of experimentation (Lopez-Rojas & Axelsson). It serves as a valuable shared resource for the research community.

The literature review highlights the effectiveness of various machine learning algorithms, the importance of feature engineering, and the innovation of synthetic data generation in advancing fraud detection research. The inclusion of "Fraud Detection on Bank Payments Using Machine Learning" further enriches the review by providing insights into practical fraud detection methods and the significance of understanding underlying patterns in credit card transactions.

Methodology

- Data Collection:

  The first step involves collecting a comprehensive dataset that contains historical transaction records from the BAF system. The dataset should include both fraudulent and non-fraudulent transactions, along with relevant features such as transaction amount, transaction type, customer information, and time of transaction. It is crucial to ensure data quality and obtain the necessary permissions to use the data. The dataset I chose, Bank Account Fraud Dataset Suite was published in NeurIPS 2022 and provided in Kaggle.

- Data Preprocessing:
  Perform data cleaning, handle missing values, and encode categorical variables in the Online Payments Fraud Detection Dataset. The first step in the methodology is to preprocess the Online Payments Fraud Detection Dataset. This involves data cleaning, handling missing values, and encoding categorical variables. Data cleaning may include removing duplicate records, correcting inconsistencies, and dealing with outliers. Missing values can be handled by imputation techniques such as mean, median, or mode imputation, or by removing records with missing values if appropriate. Categorical variables need to be encoded into numerical representations for the machine learning algorithms to process.
- Exploratory Data Analysis (EDA):
  Conduct a comprehensive EDA to gain valuable information from the dataset, understand its characteristics, and identify any patterns or anomalies associated with fraudulent transactions. It involves visualizing and analyzing the data to identify patterns, trends, and potential anomalies associated with fraudulent transactions. EDA techniques may include summary statistics, data visualization, correlation analysis, distribution analysis and time series analysis. By examining the data, we can uncover important features and gain a deeper understanding of the variables' relationships with fraud.
- Data splitting :
  Data splitting is a crucial step in machine learning model development to assess the performance and generalizability of the models. In the context of fraud detection using the preprocessed dataset, it is essential to ensure that the training and testing sets contain representative samples of both fraudulent and non-fraudulent transactions. This helps to avoid biased performance evaluations and ensures that the models can accurately classify both types of transactions.
- To maintain the class distribution and preserve the proportion of fraud instances, stratified sampling can be employed during the data splitting process. This technique helps prevent imbalanced class distributions that may negatively impact the models' ability to detect fraud accurately.
  - Identify the class labels:
  - In the preprocessed dataset, there are two class labels: fraudulent and non-fraudulent transactions.
  - Calculate the proportion of fraud instances:
  - Determine the percentage of fraud instances in the entire dataset. For example, if the dataset contains 1,000 transactions and 100 of them are fraud, the proportion of fraud instances is 10%.
  - Split the dataset:
  - Divide the dataset into training and testing sets while maintaining the class distribution.

- Ensure that both the training and testing sets have a proportion of fraud instances similar to that in the original dataset.
- Stratified sampling:
- Apply stratified sampling during the data splitting process.
- The stratified sampling technique ensures that each subset (training and testing) is representative of the overall class distribution.
- This means that the proportion of fraud instances in the training set will be similar to the original dataset, as will the proportion of non-fraudulent instances.
- Similarly, the testing set will maintain the same class distribution as the original dataset.
- By using stratified sampling, we can create training and testing sets that accurately reflect the distribution of fraudulent and non-fraudulent transactions. This approach enables fair evaluations of the models' performance in identifying fraud across different datasets.

- Model Training:
  - Implement and train the XGBoost model using the training dataset. Tune hyperparameters to optimize performance.
  - Train the SVM model using the training dataset and experiment with linear, polynomial, or radial basis function (RBF), to determine the best-performing configuration.
  - Construct an MLP classifier using the Sklearn library and train it on the training dataset. Experiment with different network architectures, activation functions, and regularization techniques to optimize performance.
  - Design and train a Multilayer Neural Network using TensorFlow/Keras, exploring different layer configurations, activation functions, and optimization algorithms to achieve optimal results.
- Model Evaluation:
  - Evaluate the trained models on the testing dataset to assess their performance.
  - Calculate the predictions for each model and compute evaluation metrics such as accuracy, precision, recall, F1-score, TPR, and FPR to measure their performance in fraud detection.
  - Generate the Receiver Operating Characteristic (ROC) curve for each model by varying the decision threshold and plotting the TPR against the FPR.
  - Compute the (AUC-ROC to quantify the models' overall discrimination ability.
- Statistical Analysis:
  - Perform statistical tests to compare the performance of the models, considering metrics such as AUC-ROC, TPR, and FPR.

    - Conduct paired t-tests or Wilcoxon signed-rank tests to determine if there are significant differences between the models' performance.
    - Calculate effect sizes, such as Cohen's d, to assess the practical significance of any performance differences observed.

- Interpretation and Conclusion:
    - Interpret the results of the performance comparison and discuss the algorithm that demonstrates superior performance in terms of the ROC curve, TPR, and FPR.
    - Analyze the strengths and limitations of each algorithm, considering factors such as model interpretability, computational complexity, and scalability.
    - Discuss the implications of the findings for fraud detection in the banking sector and provide recommendations for selecting the most effective algorithm.

Ethical challenges and Risks

One major concern centers around the potential for biased outcomes. Machine learning models that learn from biased historical data can inadvertently perpetuate these biases, leading to unfair treatment and social inequalities in fraud detection.

Transparency poses another challenge. Complex algorithms like neural networks can be difficult to interpret, eroding trust and impeding explanations, especially in financial fraud detection.There is a risk of overreliance on automation. While machine learning enhances accuracy, human oversight is vital to detect unnoticed errors and ensure fairness.

Ensuring data privacy is crucial. Safeguarding financial data and adhering to regulations are essential to prevent breaches and protect privacy. Unintended consequences are a possibility. Implementing new models may introduce unforeseen vulnerabilities. Continuous testing and monitoring are crucial to identify emerging risks. Solutions encompass explainable AI for transparency, diverse training data for reducing bias, and integrating human oversight for accountability.

Future work

I am to provide the executive summary as a written summary capturing the essence of my research project concisely, addressing the problem of financial fraud detection and the need for sophisticated, ethical solutions. My research questions will focus on evaluating the performance of machine learning algorithms within the BAF dataset, and I will outline my methodology, including undersampling and algorithmic analysis. Additionally, I will emphasize the ethical

considerations that guided my decisions. The findings, particularly the exceptional performance of the Multilayer Neural Network with TensorFlow/Keras, should be highlighted for their contribution to the advancement of fraud detection practices. Finally, my conclusion will reinforce the alignment of my findings with my initial objectives, showcasing the synthesis of theoretical understanding and practical application.

Project Timeline and Milestones

02nd June	Conducted an in-person meeting with supervisor to discuss the project idea, dissertation proposal, and introduced the ethics forms. Discussed the overall project scope and objectives.
15th June	By June 15th, 2023, planning to complete the literature review and have a draft of the project proposal ready.
19th June	meeting with my supervisor to focus on the revised dissertation proposal and literature review.
25th June	Start working on obtaining the dataset and initiating descriptive statistics and exploratory data analysis.
02nd July	I intend to complete the ethics form and send it to my supervisor for review and sign-off.
10th July	meeting with my supervisor to discuss research data, present descriptive statistics, and outline research questions.
14th July	submit the ethics form
20th July	I plan to finalize the literature review, explore similar works, and define the research method and approach.
25th July	I will begin coding and conducting the analysis phase of the project.
31st July	meeting with my supervisor to discuss research methods and showcase preliminary analysis results.
03rd August	I aim to complete the coding phase and be ready to demonstrate results to my supervisor.
10th August	final meeting with my supervisor to discuss

	project results, findings, and the report writing plan
11th August	I will focus on completing the report
16th August	Sent a draft of the project report to the supervisor for feedback
22nd August	Address the corrections and feedback highlighted by the supervisor in the draft report and finalise the report.
01st September	Submit the final report

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Appendix 2: Tableau Dashboard Link

https://public.tableau.com/views/Book1_16926601665730/Dashboard1?:language=en-US&publish=yes&:display_count=n&:origin=viz_share_link